BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BTECH
BRANCH: CSE

SEMESTER : VI/ADD
SESSION : SP/2026

SUBJECT: CS333 COMPILER DESIGN

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

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			CO	BL
Q.1(a)	Consider the following code fragment in C. The output of this program are 2 values : 13 and 16. int main() { const int i = 015; int j = 0X10; float retarn; const int* p ; int *const q = &i; printf("%d %d \n", i, j); return 0; } Identify all the distinct lexemes in the code given above and group them into the tokens, {identifier, keyword, operator, number, parentheses, delimiter, etc.}. Write the number of distinct tokens.	[2]	CO1	
Q.1(b)	Identify errors or warnings from the compiler, if any, in the code given above. If applicable, report the phase of the compiler that issues it with reasons in one line. Write regular expression for all the numbers used in the program: {015, 0X10, 0} are recognized correctly as octal, hexadecimal and decimal number.	[3]	CO2	
Q.2(a)	Consider the properties of the binary operators as shown in Table. Two or more operators in the same row are separated by a *comma*. Without using any grammar, construct a parse tree directly for the expression: 20 + 5 | | (10 != 7) && 1 <= 5, that satisfies all the properties stated above. Calculate and write the value at each node of the tree and find the value of the entire expression. Assume that 0 is *false* and any other value is *true*.	[2]	CO2	
Q.2(b)	Write an unambiguous grammar involving the operators (given in **Q.2(a)**), NUM (token for unsigned numbers), and parentheses '(' and ')' with their usual meaning.	[3]	CO3	
Q.3(a)	Construct FIRST( ) and FOLLOW ( ) for the all the non-terminals of the given grammar. This grammar is used for declaring zero or more function prototype declarations in C. There are 6 terminals { **int id ( ) ; ,** } and S is the start nonterminal. S → F S | ε F → int id (X); X → int Y | ε Y → , int Y | ε	[2]	CO3	
Q.3(b)	Construct LL(1) parsing table for the non-terminals: S, X and Y of **Q.3(a).**	[3]	CO3	
Q.4(a)	Consider the grammar: S→A = B | ε A→ id B→A You have to write a procedure for **S**, the start symbol, for a recursive descent parser. You are permitted to use the symbols ***lookahead*** to refer to the ***nexttoken*** in the input, ***error*** to indicate an error and ***success*** to refer to successful parsing. All procedures return *false* or *true* depending on whether there was an error detected in their body or not.	[2]	CO3	

Precedence	Operators	Associativity
20	| |	Right
18	&&	Left
16	!=, ==	Right
15	<, <=	Left
12	+, −	Right

PTO

Q.4(b) For a given grammar G = (N, T, P, S), two parsers, SLR(1) and LALR(1) are manually constructed. Both the parsers are free of any conflicts. Information about each parser is given below. Note that notation: |X| denotes the *number* of elements in X. [3] CO3

Parser	Action Table				goto
	|states|	|shift actions|	|reduce actions|	|error entries|	|blanks|
SLR(1)	P1	S1	R1	E1	B1
LALR(1)	P2	S2	R2	E2	B2

You have to write the relation that in general holds between {P1, P2}, {S1, S2}, {R1, R2}, {E1, E2} and {B1, B2}, justifying your answer in brief.

Q.5(a) Consider the grammar G with 3 nonterminals {S, L, R} and 3 terminals {= id *} and 6 rules : [2] CO3

1. S →L = R 2. S → R 3. R → L 4. R→ id 5. L → *R 6. L→ R

You have to construct the set of all LR(0) items for a SLR(1) parser for ONLY 3 states, (i) state 0, (ii) state s1, where s1 = **goto** (0, R), and (iii) state s2 = **goto**(0, L). For each of the specified states, write the type of each LR(0) item {kernel, incomplete, complete}, and determine whether there are any *conflicts* in the state with reasons

Q.5(b) Construct the set of all LR(1) items for a Canonical LR (1) parser for ONLY 3 states, (i) state 0, and (ii) state s3, where s3 = **goto** (0, R), and (iii) s4 = **goto**(0, L). For each of the specified states, determine whether there are any *conflicts* in the state with reasons. [3] CO3

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